INDIA ROAD ACCIDENT ANALYSIS

A Data Analytics Case Study | 2020–2024

End-to-end workflow: data cleaning, exploratory analysis, SQL analysis, and an interactive Power BI dashboard.

487,707	177,175	471,441*
Accidents in 2024	Fatalities in 2024	Sum of supplied collision categories (2024)

Prepared as a portfolio case study for the SWYNEX Technologies Data Analytics Internship.

*The injury figure is the sum of the provided collision-category values; it should not be interpreted as a verified complete national injury total.

1. Executive Summary

This case study examines supplied Indian road accident data for 2020–2024 using a practical analytics workflow. The project combines data preparation, exploratory data analysis in Excel and SQL, and an interactive Power BI dashboard. The goal is to summarize changes over time, compare states, and explore the supplied collision-category data.

2. Problem Statement and Objectives

Road-safety datasets can be difficult to interpret when figures are spread across multiple tables and years. This project organizes the supplied datasets into a consistent workflow and uses analysis and visualization to answer practical questions about accident and fatality trends.

Objectives

- Clean and prepare the supplied accident, fatality, and collision-category datasets.
- Compare annual totals and calculate year-over-year changes.
- Identify states with notable increases or decreases in reported accidents.
- Build an interactive dashboard for exploring the data.
- Document key findings, limitations, and possible next steps.

3. Dataset Overview

The project uses three cleaned datasets covering 2020–2024: accidents by state, fatalities by state, and injuries grouped by collision type. The accidents table contains state-level annual counts and year-over-year change fields. The fatalities table contains annual killed counts and a percentage-change field. The collision dataset contains category-level values such as hit and run, hit from side, hit from back, collision with parked vehicle, run off road, fixed object, vehicle overturn, head-on collision, and others.

The analysis is limited to the fields available in these datasets. The supplied data does not provide all possible explanatory variables, such as road conditions, time of day, weather, vehicle type, or detailed causal factors.

4. Data Cleaning and Preparation

The cleaning workflow preserved raw data separately and used cleaned copies for analysis. The work included removing blank rows, removing unnecessary columns, standardizing column names, checking numeric fields, and handling summary rows such as “All India” where state-level comparisons required individual states only.

- Removed blank rows and unnecessary fields from the working datasets.
- Converted annual count fields to numeric/whole-number values where required.
- Standardized SQL column names to snake_case for easier querying.
- Reviewed missing values; the accidents dataset includes unavailable values for Ladakh.
- Kept the supplied collision-category totals distinct from the national accident and fatality totals.

5. Exploratory Data Analysis

5.1 Annual accident totals

Year	Reported accidents
2020	372,181
Year	**Reported accidents**
2021	412,432
2022	461,312

2023	480,583
2024	487,707

Reported accidents increased from 372,181 in 2020 to 487,707 in 2024, an increase of about 31.04% over the period. Year-over-year growth slowed from 11.85% in 2022 to 4.18% in 2023 and 1.48% in 2024.

5.2 Annual fatalities totals

Year	Reported fatalities
2020	138,383
2021	153,972
2022	168,491
2023	172,890
2024	177,175

Reported fatalities rose from 138,383 in 2020 to 177,175 in 2024. Between 2023 and 2024, fatalities increased by approximately 2.48%, which is higher than the 1.48% growth in reported accidents over the same interval.

5.3 State-level changes in accidents (2023–2024)

Largest increases among the states in the analysis: Telangana (+3,083; +13.5%), Uttar Pradesh (+1,518; +3.4%), Chhattisgarh (+1,389; +10.3%), Madhya Pradesh (+1,342; +2.4%), and Maharashtra (+875; +2.5%).

Largest decreases among the states in the analysis: Gujarat (−761; −4.7%), Haryana (−657; −6.3%), Jammu & Kashmir (−490; −7.8%), Andhra Pradesh (−392; −2.0%), and Karnataka (−378; −0.9%).

5.4 Collision-category analysis

The supplied collision-category dataset reports a combined value of 462,825 for 2023 and 471,441 for 2024, an increase of approximately 1.86%. Because this is a sum of the categories provided, it is presented as a collision-category total rather than a verified complete national injury total.

6. SQL Analysis

MySQL was used to inspect the cleaned accidents data, calculate annual totals, compare states, and summarize year-over-year changes. The SQL workflow supports reproducibility by keeping analytical steps in queries rather than relying only on manual calculations.

Examples of analysis questions addressed:

- What is the annual total of reported accidents from 2020 through 2024?
- Which states have the highest accident counts in 2024?
- Which states experienced the largest increases or decreases from 2023 to 2024?
- How did annual accident growth change over the five-year period?

7. Interactive Power BI Dashboard

The dashboard brings the analysis into one interactive view. It includes KPI cards, state-wise accident comparisons, collision-category visuals, a fatalities trend line for 2020–2024, and slicers for State and Collision

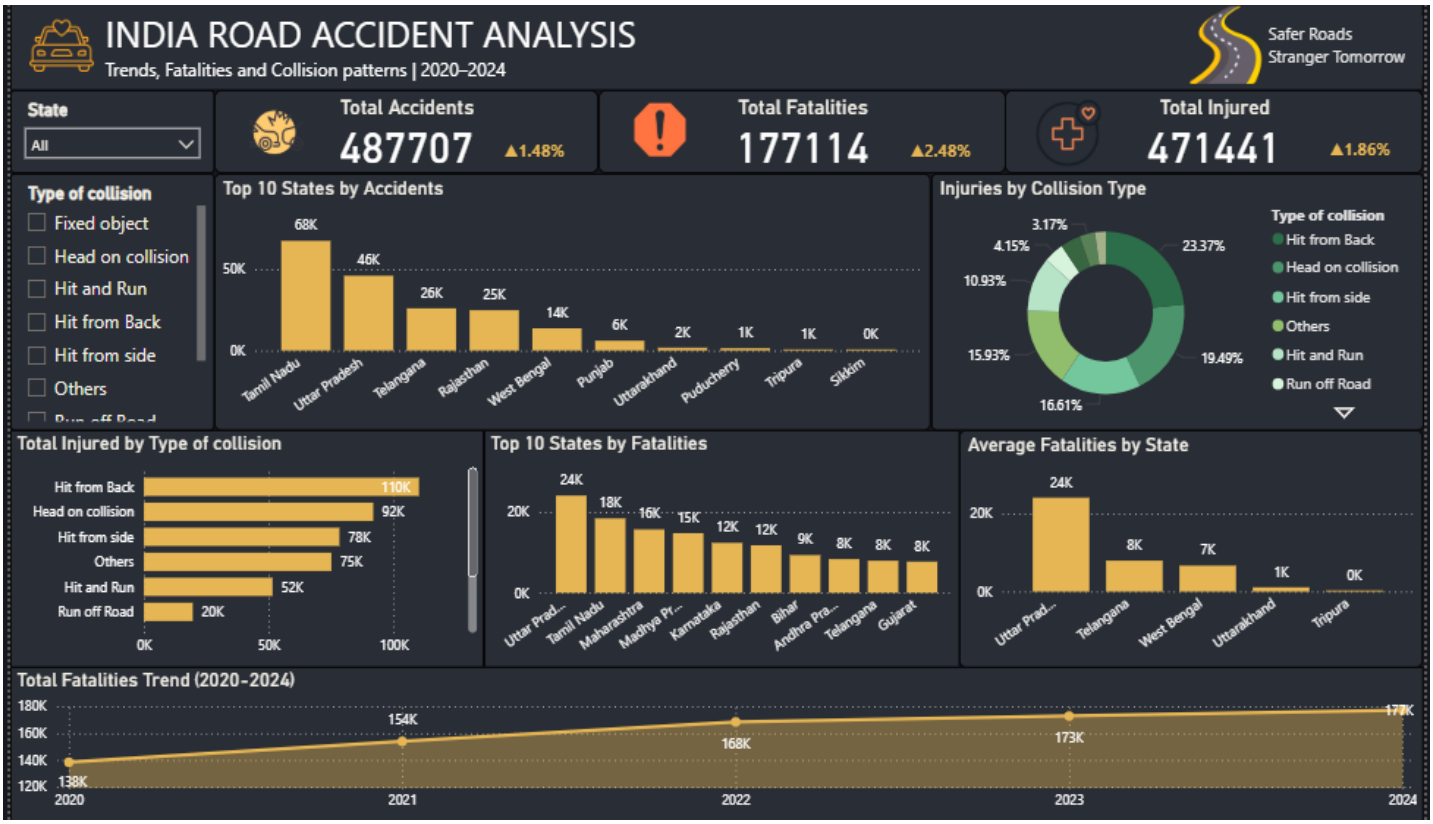
Type. The State slicer is connected to the fatalities trend table so the line chart can display a selected state's trend.

Dashboard KPIs currently used in the project:

• 2024 reported accidents: 487,707

• 2024 reported fatalities: 177,175

• 2024 supplied collision-category total: 471,441 (not a verified complete national injury total)



The Power BI dashboard provides an interactive overview of road accidents in India from 2020 to 2024. It includes KPI cards, state-wise accident comparisons, collision category analysis, and fatalities trend chart. State and collision-type slicers allow users to explore the available data interactively.

8. Key Findings

- Reported accident counts increased each year in the supplied 2020–2024 series.
- The accident growth rate slowed to 1.48% in 2024, after faster growth in 2021 and 2022.
- Reported fatalities continued to rise and increased by approximately 2.48% from 2023 to 2024.
- State-level patterns varied: Telangana recorded the largest absolute increase among the listed states, while Gujarat recorded the largest decrease.
- Collision-category values rose by approximately 1.86% from 2023 to 2024 in the supplied dataset.

9. Recommendations

- Use state-level changes to identify locations for deeper road-safety investigation, rather than assuming the count change explains its cause.
- Combine these summary datasets with road type, time of day, weather, vehicle type, and cause-of-accident data in future analysis.
- Validate definitions and coverage across datasets before comparing totals, especially when collision categories may not cover every injury record.

- Keep the Power BI slicers and chart labels clear so users can explore patterns without misreading totals or percentages.

10. Limitations

- The analysis is based only on the fields present in the supplied datasets.
- The available summaries do not explain causal factors behind changes in accident or fatality counts.
- Ladakh has unavailable values in the accidents dataset; missing data should not be interpreted as zero.
- The supplied collision-category sum is not treated as a verified national injury total.
- State comparisons use reported counts and do not adjust for population, vehicle registrations, traffic exposure, or travel distance.

11. Conclusion

This project demonstrates an end-to-end data analytics workflow: preparing datasets, exploring trends with Excel and SQL, and communicating results through an interactive Power BI dashboard. The analysis shows that reported accidents and fatalities increased over the period covered by the supplied data, while the annual growth rate of accidents slowed in 2024. Further analysis with richer contextual and causal data would be needed to explain these patterns and support targeted interventions.

12. Tools and Deliverables

Tools: Microsoft Excel, MySQL Workbench, Microsoft Power BI, Power Query, and DAX.

Deliverables: Cleaned CSV datasets, SQL analysis queries, Power BI dashboard (.pbix), GitHub repository, and this case study report.

GitHub repository: https://github.com/akhilareddybattula/SWYNEX-Data-Analytics-Internship